

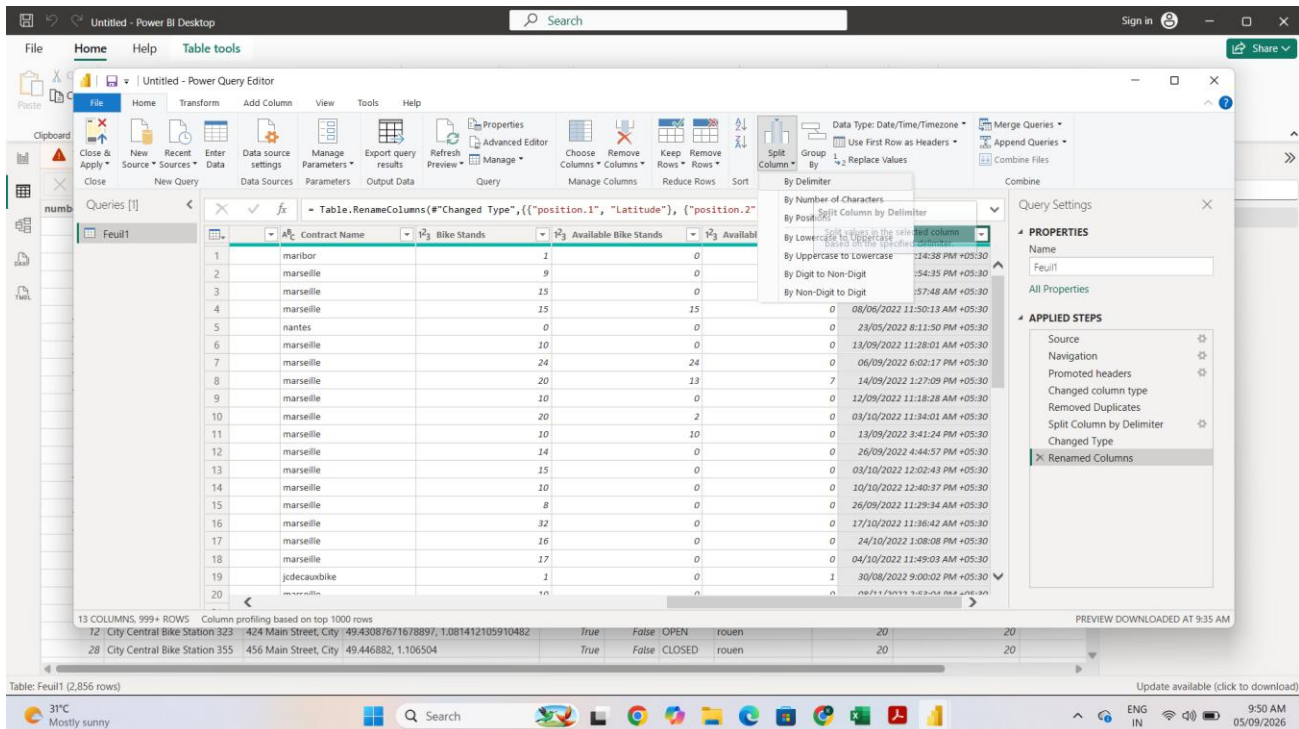
Data Transformation, Modelling and Visualization with Power BI

1. Getting the Data

The workflow begin with importing the source data into Power BI using Home → Get data

2. Data Transformation – Power Query

2.1 Prepare the source table and clean columns



The Power Query Editor shows the Feuil1 query with the source data loaded. The Applied Steps pane records Source, Navigation, Promoted Headers, Changed Column Type, Removed Duplicates, Split Column by Delimiter, Changed Type, and Renamed Columns. The Split Column menu is open, indicating that a text field is being separated using a delimiter.

2.2 Split a column using a comma delimiter

Split Column by Delimiter

Specify the delimiter used to split the text column.

Select or enter delimiter

Comma

Split at

☒ Left-most delimiter

☐ Right-most delimiter

☐ Each occurrence of the delimiter

Advanced options

Quote Character

' '

☐ Split using special characters

Insert special character

OK Cancel

Query Settings

PROPERTIES

Name

Feu1

APPLIED STEPS

Source

Navigation

Promoted headers

Changed column type

Removed Duplicates

Split Column by Delimiter

Changed Type

The Split Column by Delimiter dialog is open in Power Query. The delimiter is set to Comma and the split option is set to the left-most delimiter. This is used to separate combined text values into distinct columns for Latitude and Longitude

2.3 Rename and type geographic/station fields

Table.RenameColumns(#'Changed Type',({('position.1', 'Latitude'), ('position.2', 'Longitude')}))

Query Settings

PROPERTIES

Name

Feu1

APPLIED STEPS

Source

Navigation

Promoted headers

Changed column type

Removed Duplicates

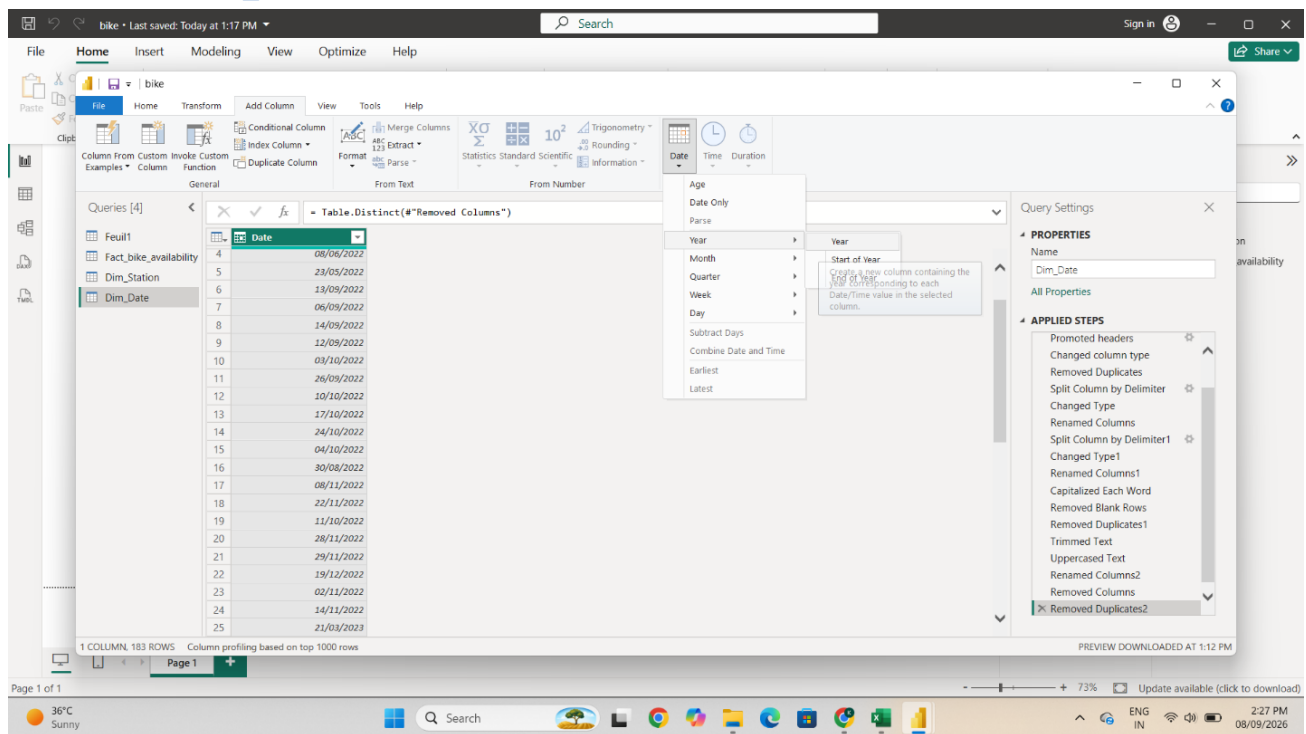
Split Column by Delimiter

Changed Type

Renamed Columns

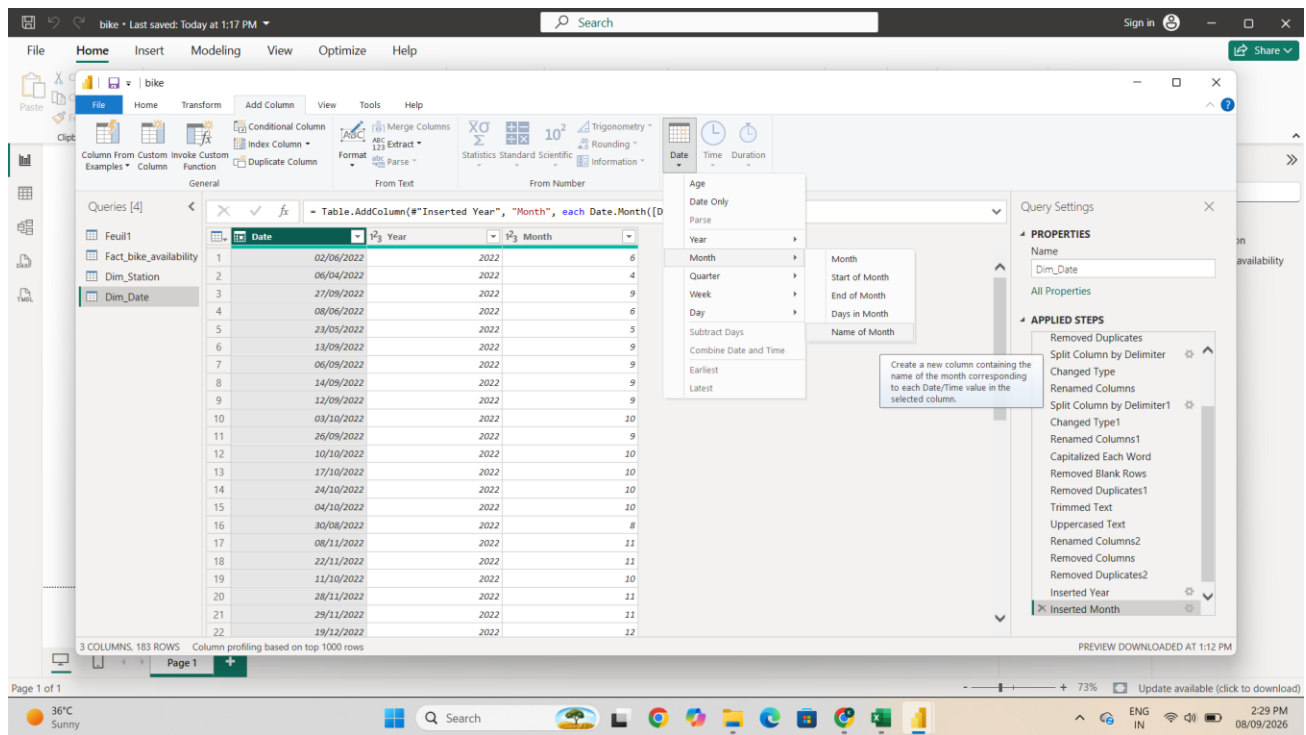
The transformed table displays fields such as name, address, Latitude, Longitude, banking, and bonus. The formula bar shows a Rename Columns operation, and the Applied Steps pane includes Changed Type and Renamed Columns.

2.4 Create the Dim_Date base table



The Dim_Date query is selected in Power Query. The table contains a Date column with unique dates, and the Applied Steps show cleaning actions including Removed Duplicates. The Date menu is opened to derive date attributes such as Year.

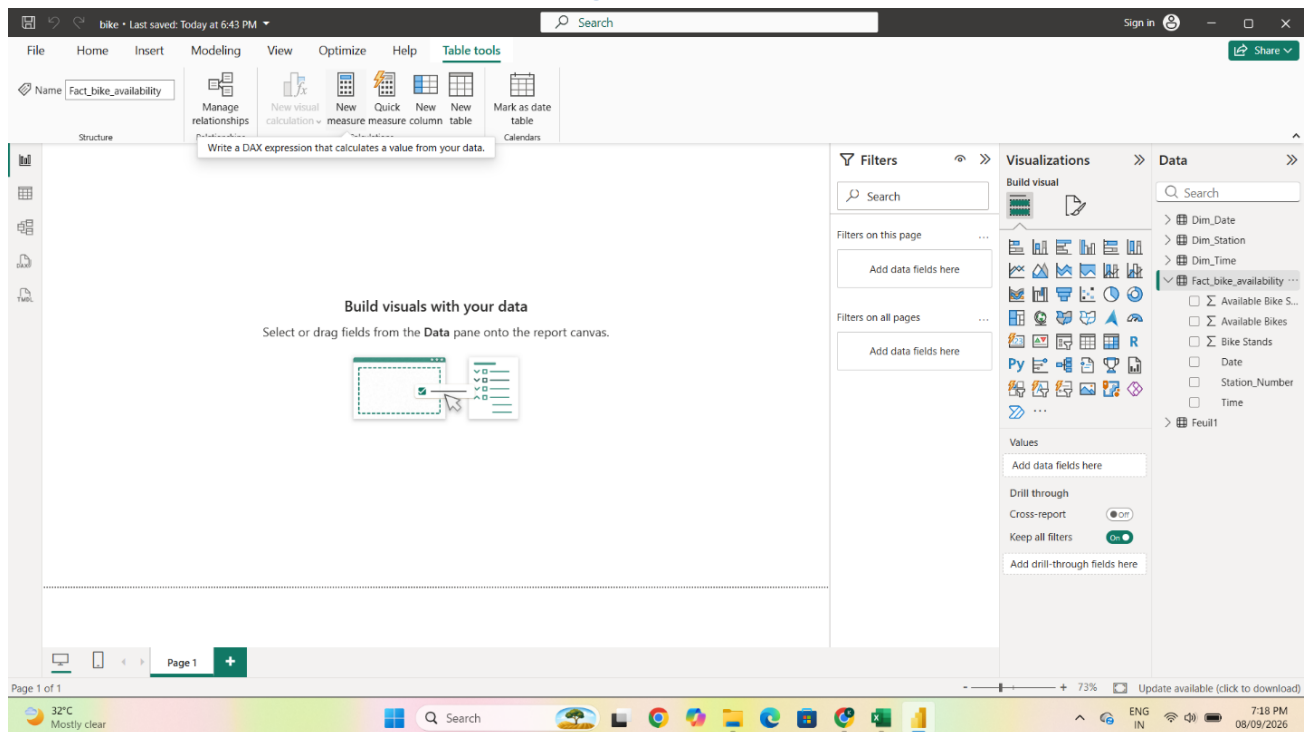
2.5 Add Month to Dim_Date



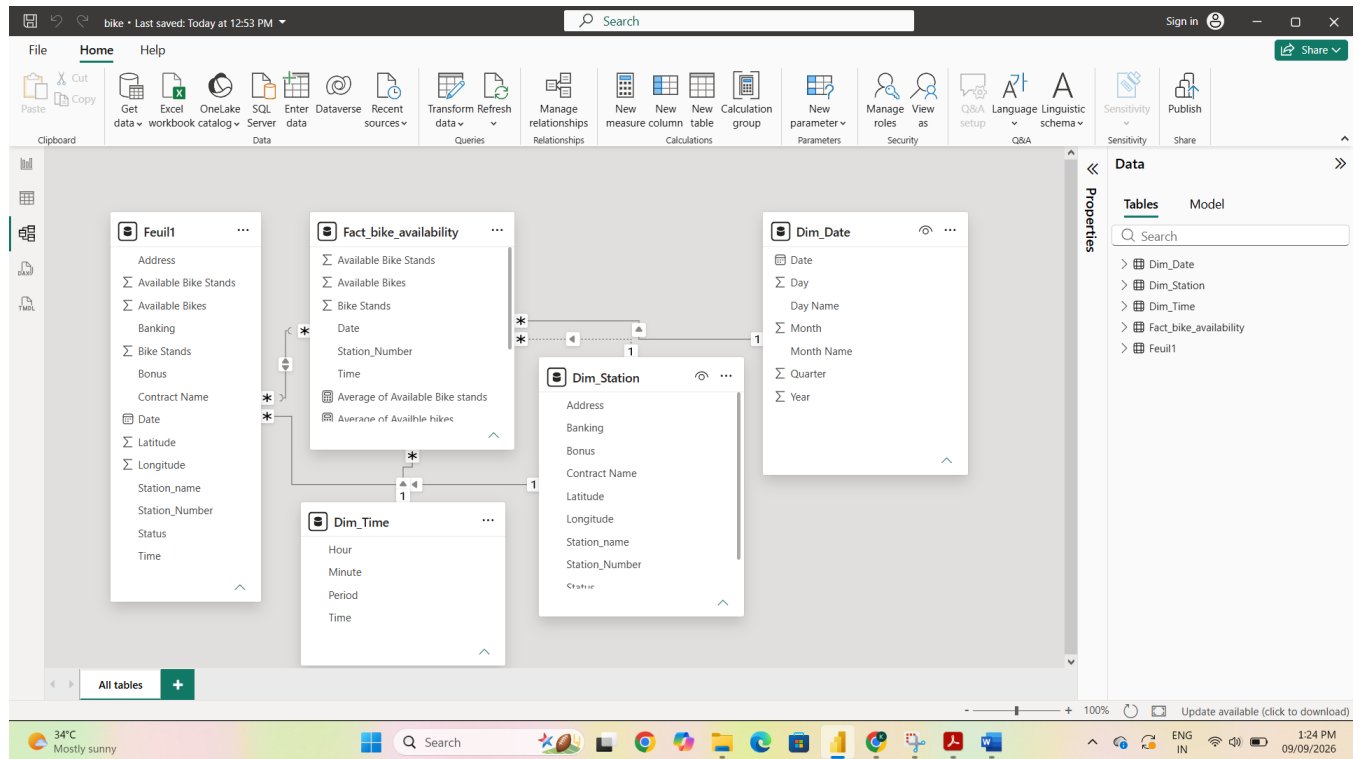
The Dim_Date query now contains Date, Year, and Month columns. The Date transformation menu is open on the Month options, and the Applied Steps pane shows an Inserted Month step.

3. Data Modelling

3.1 Confirm the model tables before creating measures

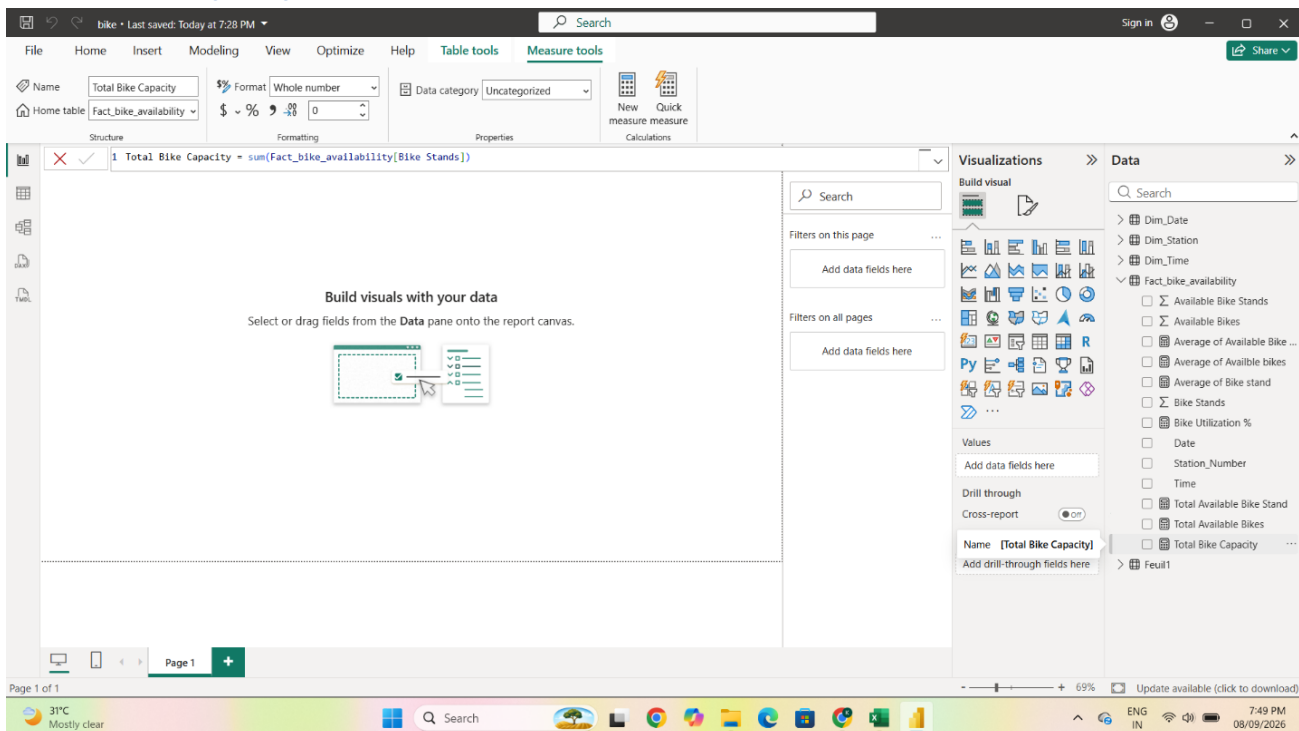


The Data pane containing Dim_Date, Dim_Station, Dim_Time, Fact_bike_availability, and Feuil1. The selected table is Fact_bike_availability, which contains fields such as Available Bike Stands, Available Bikes, Bike Stands, Date, Station_Number, and Time.



4. DAX Measures

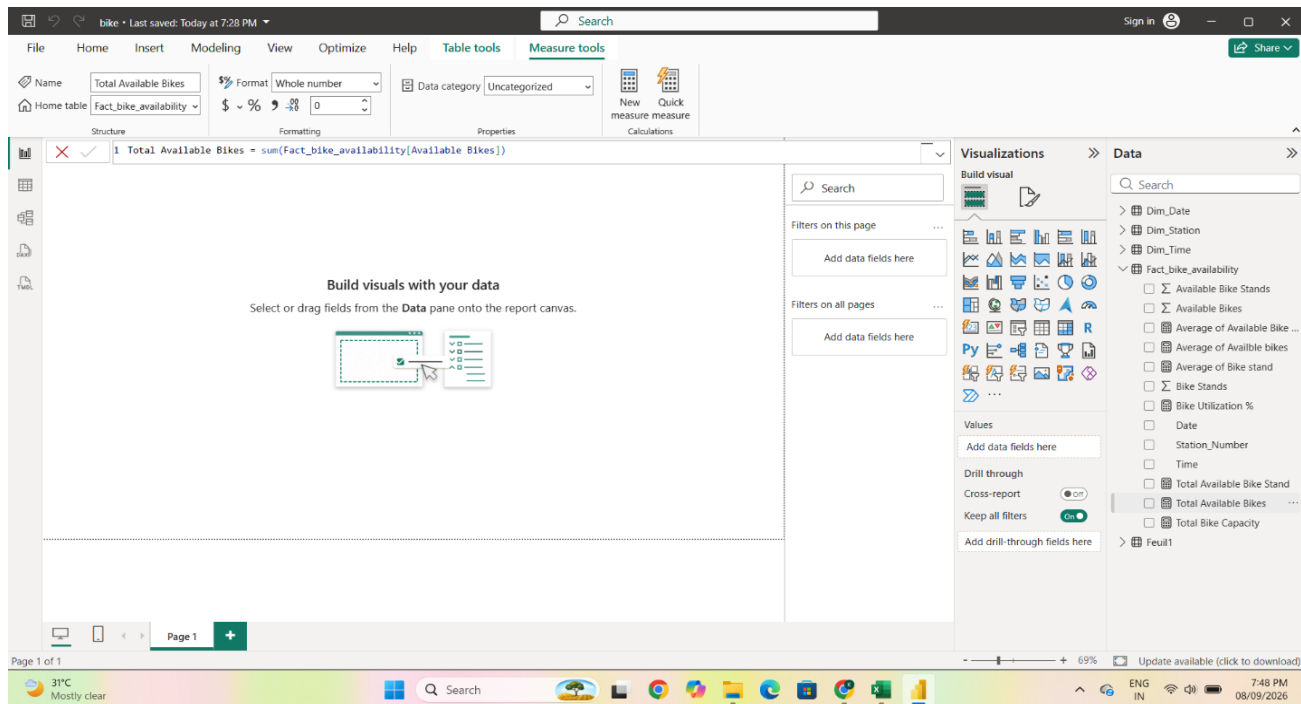
4.1 Total Bike Capacity



The measure Total Bike Capacity is created in the Fact_bike_availability table using SUM over the Bike Stands column. This measure represents the total bike capacity available across the fact data.

- DAX shown: Total Bike Capacity = SUM(Fact_bike_availability[Bike Stands])

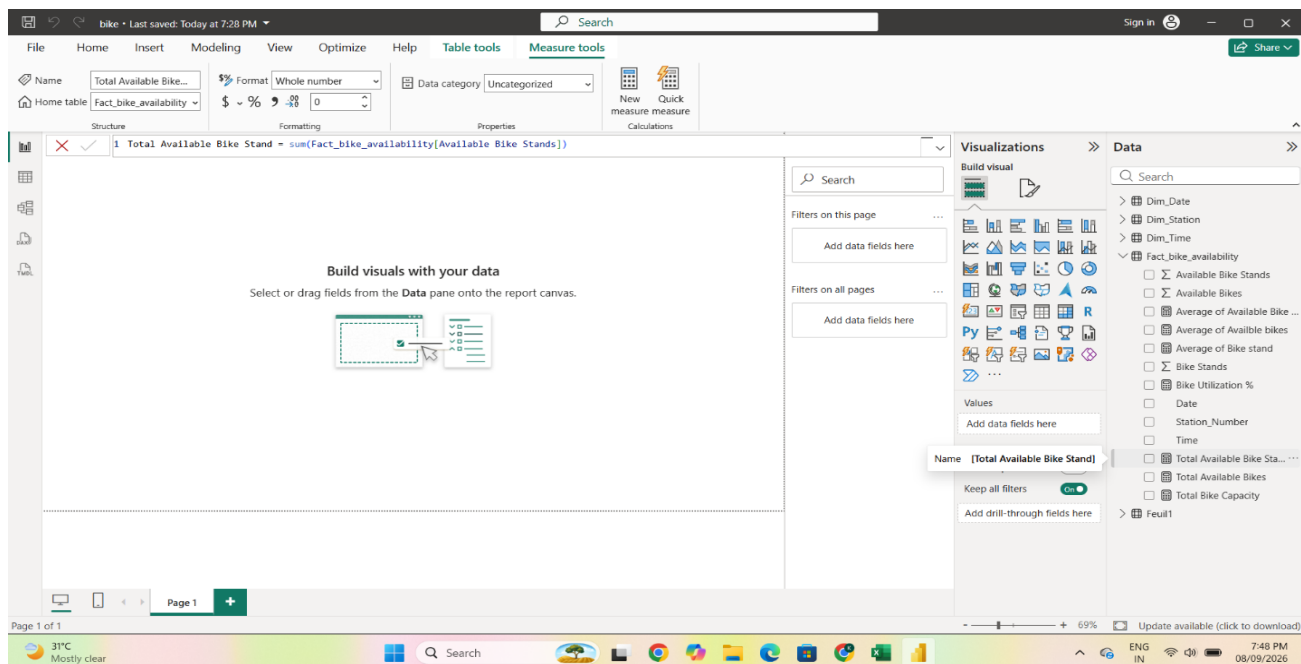
4.2 Total Available Bikes



The Total Available Bikes measure sums the Available Bikes column from the fact table. It is useful for analysing the total number of bikes available under the selected filter context.

- DAX shown: Total Available Bikes = SUM(Fact_bike_availability[Available Bikes])

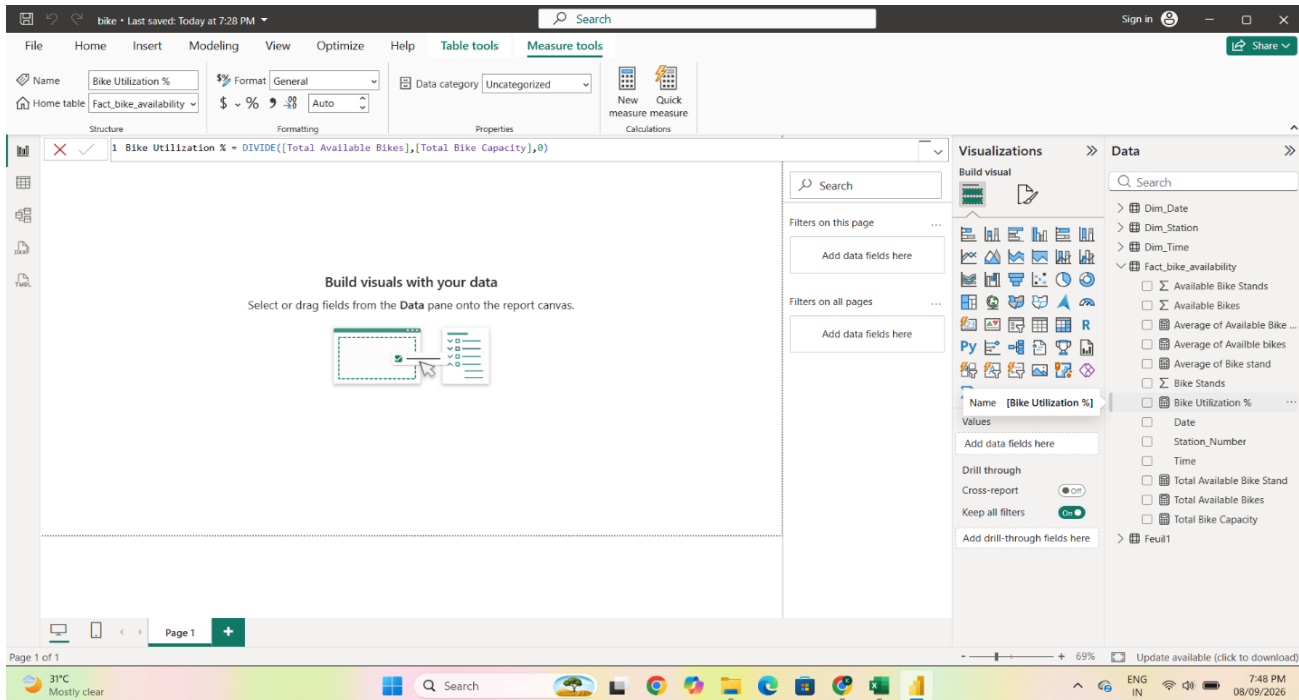
4.3 Total Available Bike Stands



The Total Available Bike Stand measure sums the Available Bike Stands column. It can be used to compare available docking capacity across stations or time periods.

- DAX shown: Total Available Bike Stand = SUM(Fact_bike_availability[Available Bike Stands])

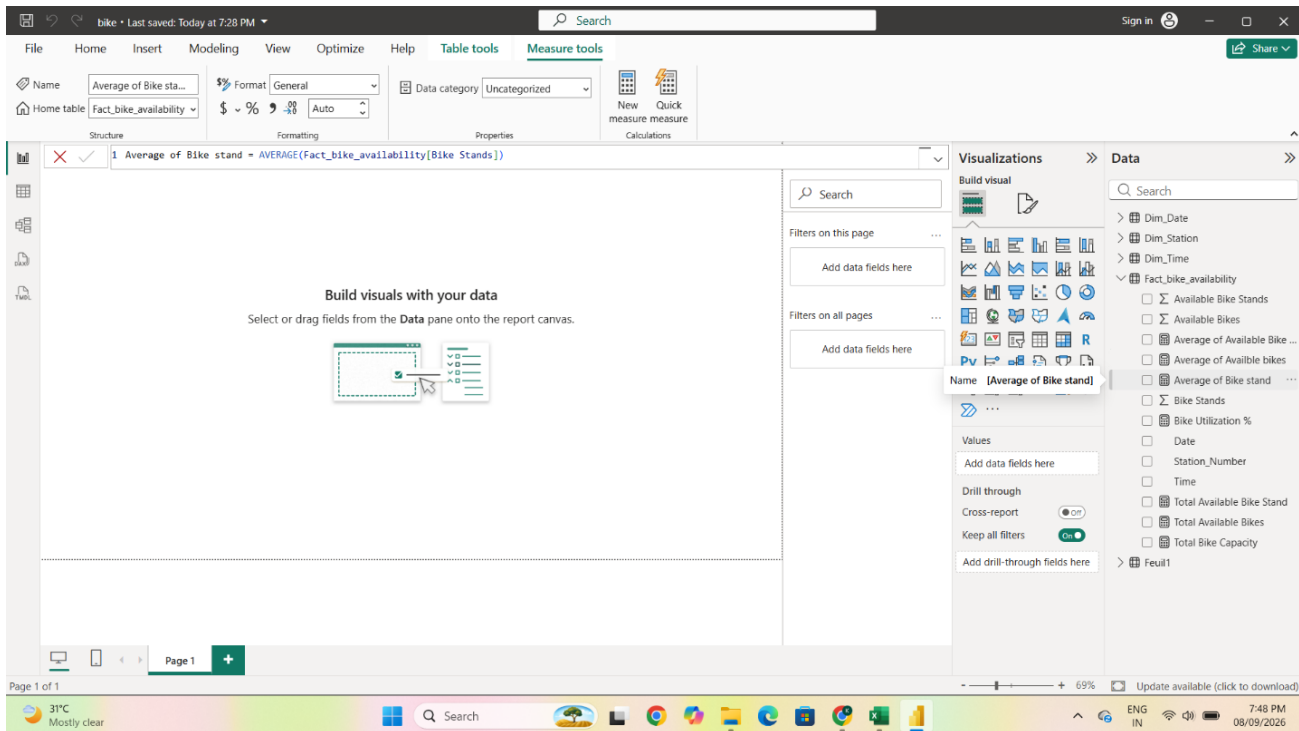
4.4 Bike Utilization %



The Bike Utilization % measure is calculated by dividing Total Available Bikes by Total Bike Capacity with DIVIDE. The third argument is 0, which provides a safe result when the denominator is zero.

- DAX shown: Bike Utilization % = DIVIDE([Total Available Bikes],[Total Bike Capacity],0)

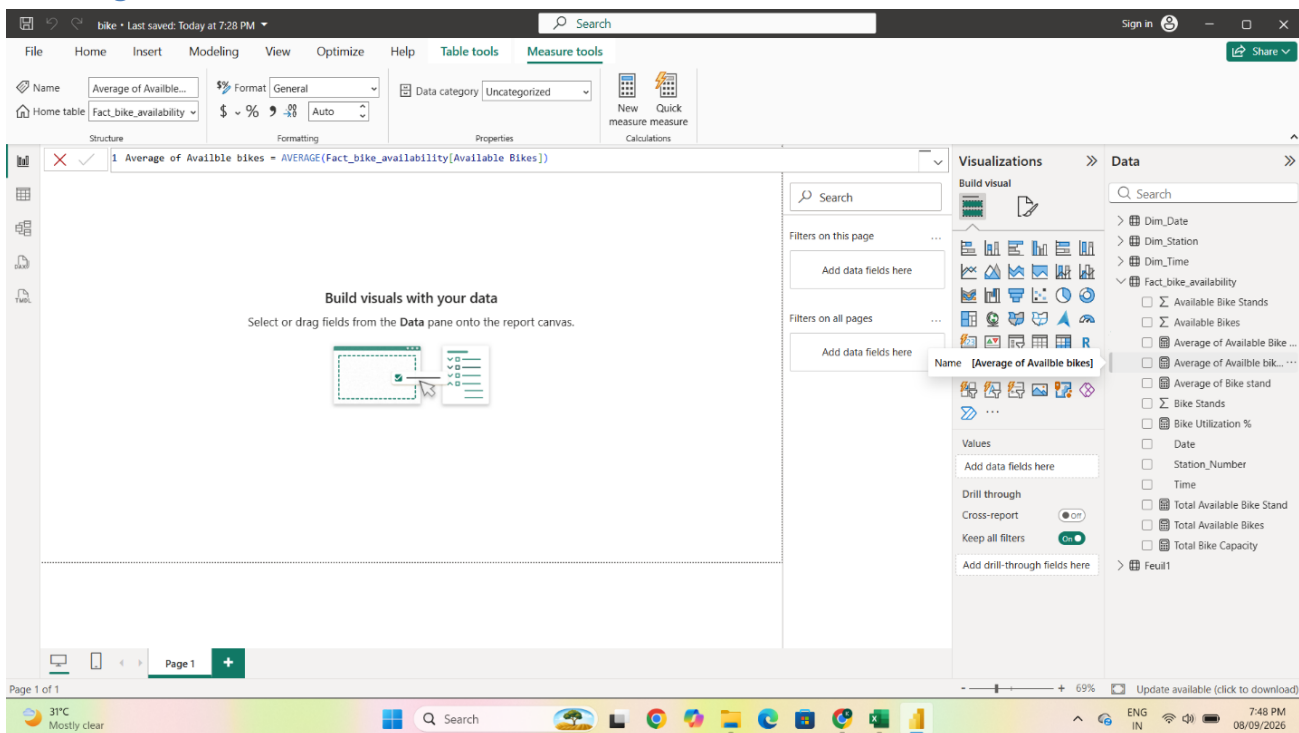
4.5 Average of Bike Stand



This measure calculates the average number of Bike Stands in the fact table using AVERAGE. It provides an average station capacity measure rather than a total.

- DAX shown: Average of Bike stand = AVERAGE(Fact_bike_availability[Bike Stands])

4.6 Average of Available Bikes



This measure calculates the average number of Available Bikes using the AVERAGE function. It can be used to study typical bike availability across the selected data.

- DAX shown: Average of Available bikes = AVERAGE(Fact_bike_availability[Available Bikes])

4.7 Average of Available Bike Stands

The screenshot shows the Microsoft Power BI Desktop interface. The 'Name' field is set to 'Average of Available Bike stands' and the 'Home table' is 'Fact_bike_availability'. The DAX formula bar contains the measure: `AVERAGE(Fact_bike_availability[Available Bike Stands])`. The 'Visualizations' pane on the right shows a list of fields, including 'Average of Available Bike stands' which has been added to the 'Values' section. The 'Data' pane on the right shows the 'Fact_bike_availability' table with fields like 'Available Bike Stands', 'Average of Available bikes', 'Average of Available Bike stand', 'Bike Stands', 'Bike Utilization %', 'Date', 'Station_Number', 'Time', 'Total Available Bike Stand', 'Total Available Bikes', and 'Total Bike Capacity'.

This measure calculates the average number of Available Bike Stands using AVERAGE. It complements the total available stand measure by showing the typical level of available stand capacity.

- DAX shown: Average of Available Bike stands = AVERAGE(Fact_bike_availability[Available Bike Stands])

4.8 Number of Stations

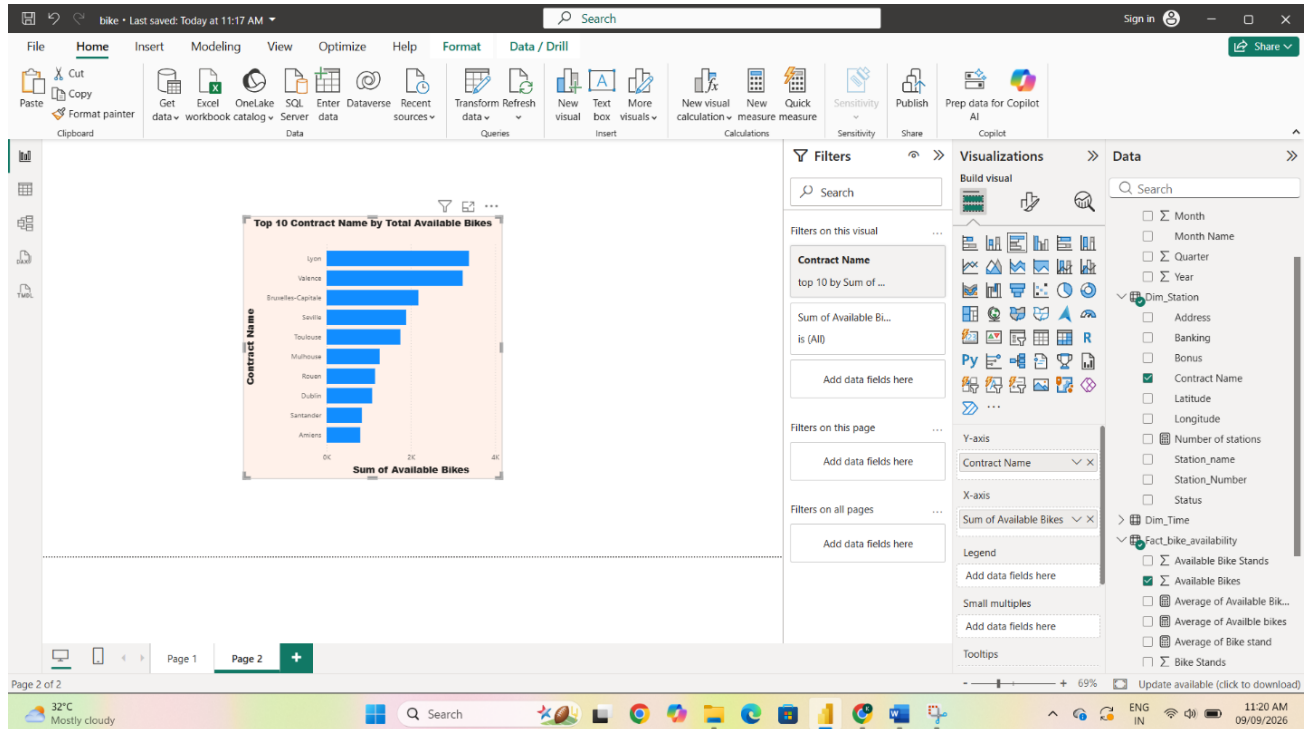
The screenshot shows the Microsoft Power BI Desktop interface. The 'Name' field is set to 'Number of stations' and the 'Home table' is 'Dim_Station'. The DAX formula bar contains the measure: `DISTINCTCOUNT(Dim_Station[Station_Number])`. The 'Visualizations' pane on the right shows a list of fields, including 'Number of stations' which has been added to the 'Values' section. The 'Data' pane on the right shows the 'Dim_Station' table with fields like 'Address', 'Banking', 'Bonus', 'Contract Name', 'Latitude', 'Longitude', 'Number of stations', 'Station_name', 'Station_Number', 'Status', 'Dim_Time', 'Fact_bike_availability', and 'Feuil1'.

The Number of stations measure uses DISTINCTCOUNT on Dim_Station[Station_Number]. This counts unique station numbers rather than counting every row in the fact table.

- DAX shown: Number of stations = DISTINCTCOUNT(Dim_Station[Station_Number])

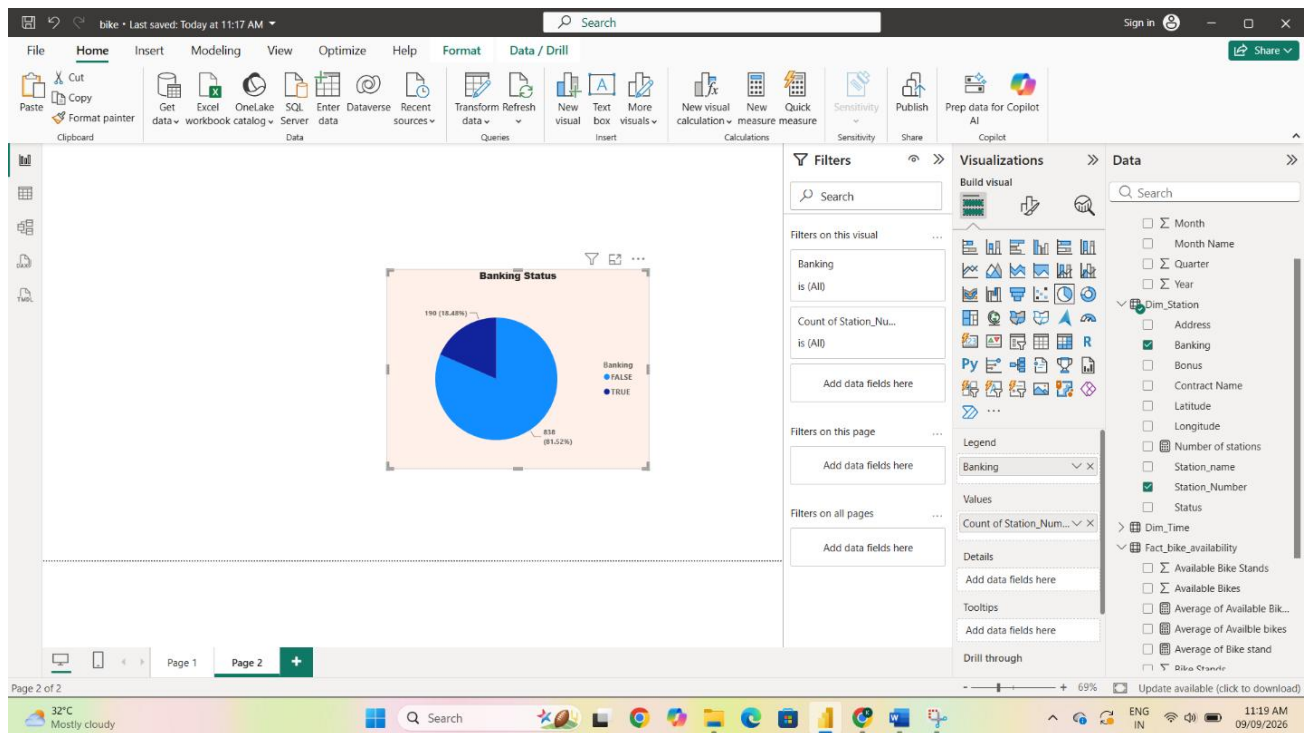
5. Charts and Visualizations

5.1 Top 10 Contract Names by Total Available Bikes



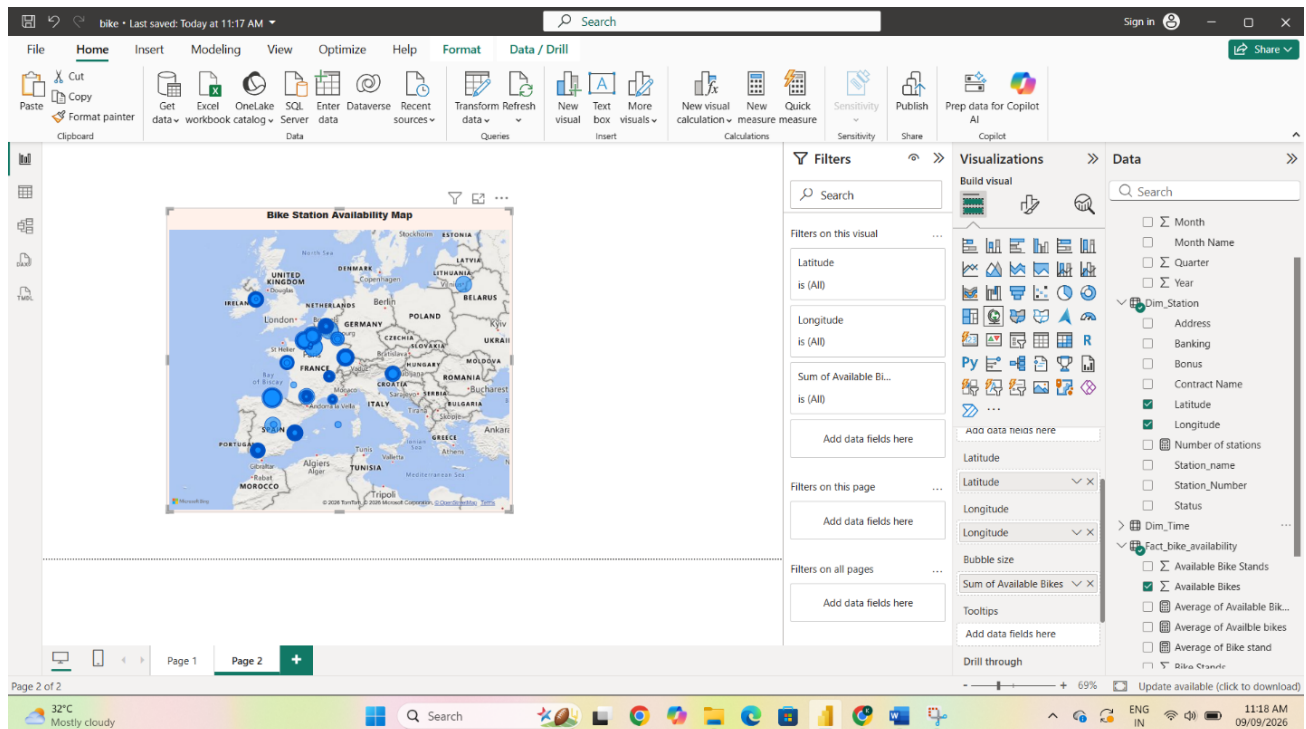
A horizontal bar chart ranks the top 10 Contract Name values by Sum of Available Bikes. A Top N visual filter is applied to Contract Name, making the chart useful for comparing the highest-volume contract locations.

5.2 Banking Status distribution



A pie chart displays the distribution of stations by Banking status. The visual uses Banking as the legend and Count of Station_Number as the value.

5.3 Bike Station Availability Map



A map visual plots bike station locations using Latitude and Longitude. The bubble size represents Sum of Available Bikes, allowing geographic comparison of bike availability.

Bike Sharing Operations & Availability Analysis

Total Available Bike Stand by Status

Status	Count	Percentage
OPEN	27K	84.1%
CLOSED	5K	15.85%

Total Bike Capacity by Year

Year	Total Bike Capacity
2022	~2K
2023	~2K
2024	~55K

Total Bike Capacity by Period

Period	Count	Percentage
Afternoon	53K	93.44%
Morning	2K	3.38%
Evening	-	-
Night	-	-

Availability by Hour

Hour	Sum of Available Bikes
0	~0K
15	20K
20	~0K

Banking Status

Banking Status	Count	Percentage
FALSE	838	81%
TRUE	190	18.4%

Top 10 Contract Name by Total Available Bikes

Contract Name	Sum of Available Bikes
Lyon	~4.5K
Valence	~4.2K
Bruxelles-Capitale	~3.8K
Seville	~3.5K
Toulouse	~3.2K
Mulhouse	~2.8K
Rouen	~2.5K
Dublin	~2.2K
Santander	~2.0K
Amiens	~1.8K

Bike Station Availability Map

Average Metrics Summary

- 32K Total Available Bike Stand
- 57K Total Bike Capacity
- 20K Total Available Bikes
- 11.25 Average of Available Bik...
- 6.86 Average of Aivble bikes.
- 20.02 Average of Bike stand
- 0.34 Bike Utilization %

Quarter Selection

Quarter:

Range slider: